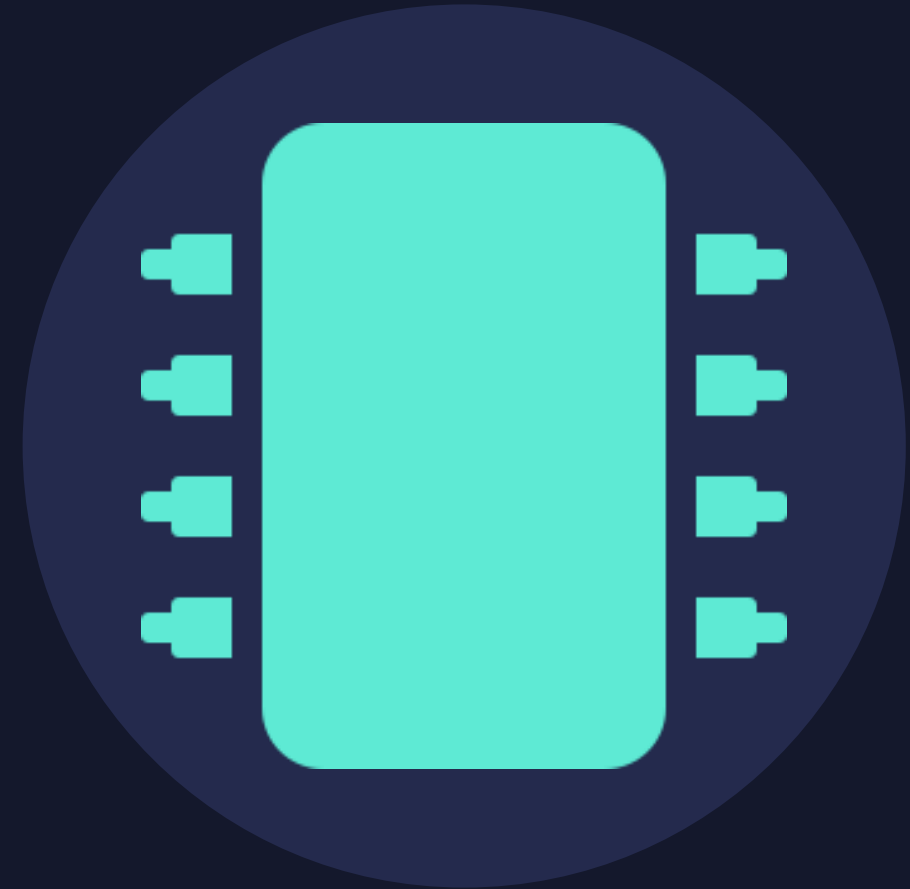


CHAPTER 1

Computer Organization and Architecture

What Is a Computer, Really?

Big ideas edition, built for discussion, not for reading along.



Architecture vs. Organization



Architecture

What the programmer sees. Instructions, registers, data types, how memory is addressed.



Organization

How it is actually built. Circuits, control signals, the hardware details a programmer never sees.

Real example: the IBM System/370 is one architecture that spanned decades of different models, each with its own organization, so old software kept running on new hardware.

Every Computer Is a Hierarchy



We zoom in one level at a time. At each level we ask two questions:

Structure -- how do the parts connect?

Function -- what does each part do?

Four Things Every Computer Does



Process

Turn data into something new.



Store

Hold data, short term and long term.



Move

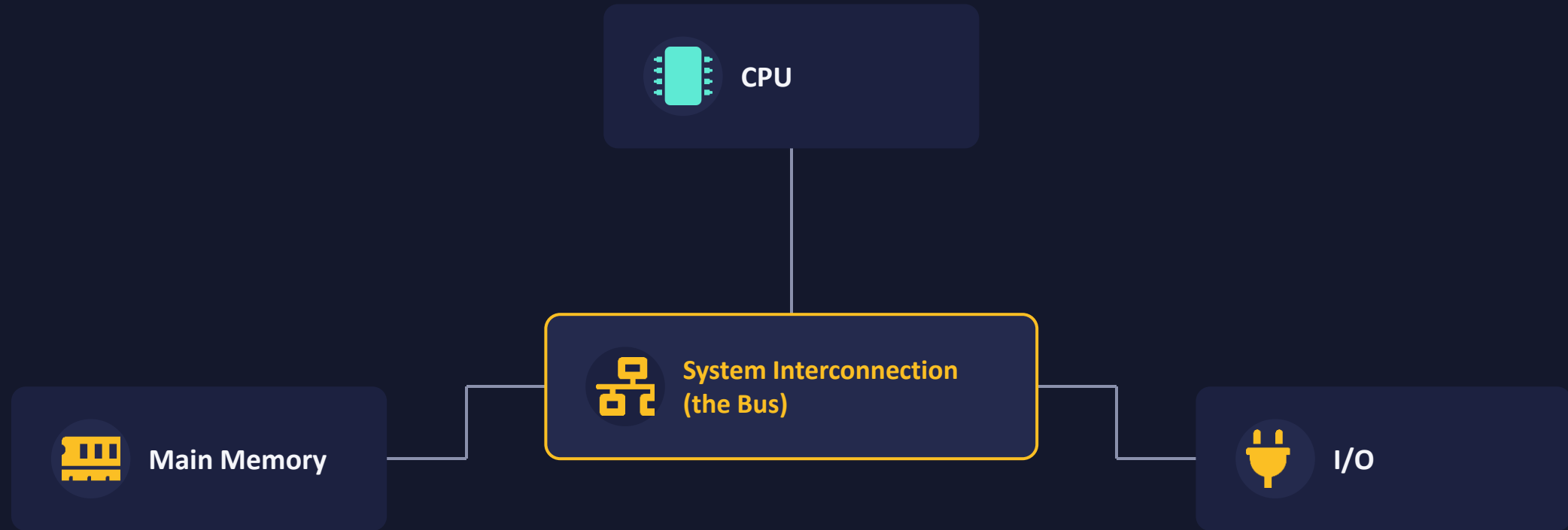
Get data in and out (I/O), or across distance (communications).



Control

Orchestrate everything else in response to instructions.

Four Building Blocks



These four blocks, wired together, are what you will go find inside a real machine this afternoon.

Open Up the CPU



Control Unit

Directs the operation of the CPU, and so the computer.



ALU

Performs the math and the logic.



Registers

Tiny, very fast storage inside the CPU itself.

Zoom in again: a modern chip usually holds many **cores**. A core is basically a mini CPU. The **processor** is the physical chip that holds one or more cores.

Cache: The Space Between Fast and Big



Closer to the core costs more per byte, so we use a little bit of very fast memory and a lot of slower memory, and let cache guess what you need next.



THINK ABOUT IT

**If cache memory is faster, why do we not just
build ALL memory that fast?**

One Idea Behind Every Computer in This Room



1945

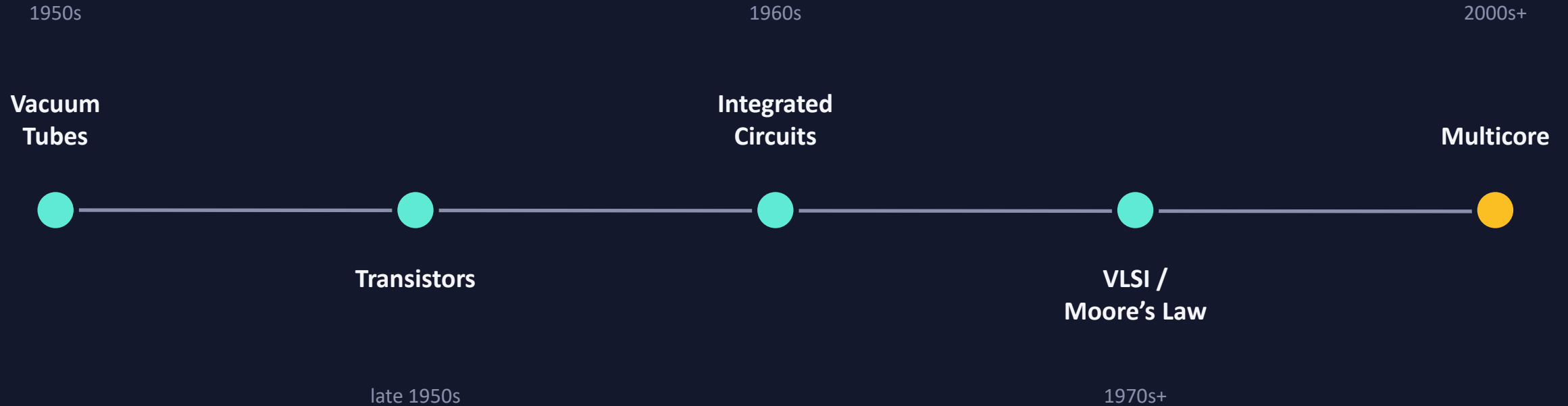
John von Neumann proposes the stored program concept in a design paper for the EDVAC.

1952

The IAS computer is completed at Princeton, the prototype for every general purpose computer since.

The big idea: instructions and data live together, in the same memory. That is what "stored program" means, and it is exactly what happens when you flash an operating system onto an SD card.

Five Generations in Seventy Years



Each generation made computers smaller, cheaper, and faster, which is exactly why a Raspberry Pi today outperforms a room sized machine from the 1950s.

Chapter 1, Meet the Raspberry Pi



Embedded systems put a full computer inside another device, like a switch, a car, or a thermostat.



The Internet of Things connects billions of these small, embedded computers together.



Most of them, including the Pi in front of you, run on ARM architecture instead of the x86 architecture found in most servers and laptops.

Same four functions. Same four building blocks. Different organization.

NOW LET'S OPEN SOME HARDWARE

From Slides to Hands On Hardware



Take It Apart

Open a server blade and a switch. Find the CPU, memory, I/O, and interconnection.



Compare

Set the Pi next to the blade. Same four jobs, very different organization.



Preview Linux

Watch what it means to write an operating system into memory, the stored program idea in action.